

Due: **September 18th, 2024** (@end of the day)

Name: _____ CMSID: _____

Instructions

1. You can solve on the PDF or print this document and then use a **pen or pencil** to write your solution.
2. Please note that the submitted work must represent your own thinking and effort. **Copying the work of others will not be tolerated.**
3. Please submit within the due time (see above). **Late submissions are not accepted.**

Answer the following:

- 1) What is the negation of this proposition?

“A sufficient condition for the sky is not clear is I will not go for a walk”

- a) I will go for a walk and the sky is not clear
- b) I will not go for a walk and the sky is clear
- c) The sky is not clear or I will not go for a walk.
- d) The sky is not clear or I will go for a walk.
- e) All of the answers are incorrect.

- 2) Let p and q be the propositions “You have a driver’s license,” and “You are at least 18 years old,” respectively. Express the following compound proposition as an English sentence:

$$q \rightarrow \neg p$$

- a) *“If you are at least 18 years old, then you have a driver’s license”*
- b) *“If you are not at least 18 years old, then you don’t have a driver’s license”*
- c) A necessary condition for “you don’t have a driver’s license” is “you are at least 18 years old.”
- d) A necessary condition for “you are at least 18 years old” is “you don’t have a driver’s license.”
- e) None of the answers are correct

- 3) Let p and q be the propositions “You have a driver’s license,” and “You are at least 18 years old,” respectively. Express the following compound proposition as an English sentence: $\neg p \wedge q$

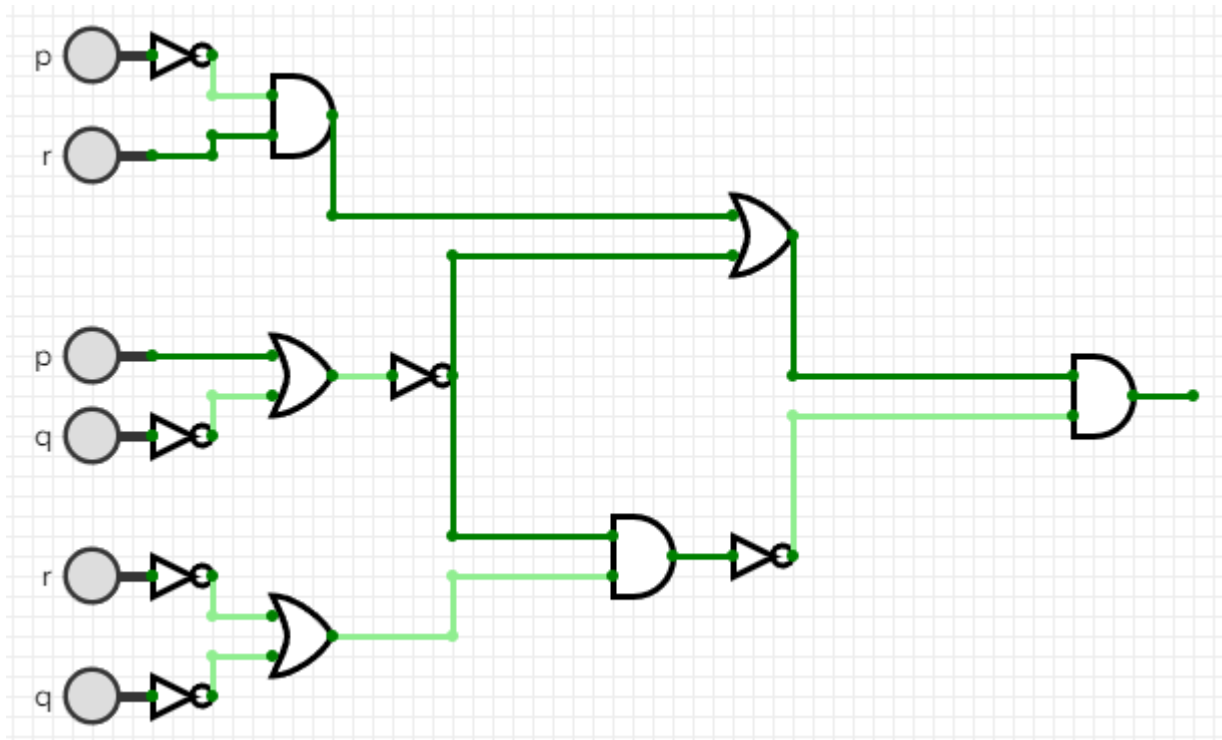
- a) *You have a driver’s license and you are at least 18 years old*
- b) *You don’t have a driver’s license and you are not at least 18 years old*
- c) *You don’t have a driver’s license or you are at least 18 years old*
- d) *You have a driver’s license or you are not at least 18 years old*
- e) None of the answers is correct

- 4) Let p , q , and r be the following propositions: p : "It is warm this afternoon", q : "it is the weekend", and r : "We will have a party outside.". Write the following statement using formula including p , q , and r and logical connectives.

"We will have a party outside if and only if it is warm this afternoon and it is the weekend"

- a) $\neg r \wedge p \wedge q$
 - b) $r \rightarrow p \wedge q$
 - c) $p \leftrightarrow q \wedge r$
 - d) $p \wedge q \leftrightarrow r$
 - e) None of the answers is correct
- 5) What is the contrapositive of the conditional statement " *I don't sleep well at night whenever I had coffee in the evening*"
- a) *If I don't sleep well at night, then I had coffee in the evening.*
 - b) *If I sleep well at night, then I had coffee in the evening.*
 - c) *If I sleep well at night, then I didn't have coffee in the evening.*
 - d) *If I don't sleep well at night, then I didn't have coffee in the evening.*
 - e) *None of the answers is correct*
- 6) What is the converse of the conditional statement " *I don't sleep well at night whenever I had coffee in the evening*"
- a) *If I sleep well at night, then I had coffee in the evening.*
 - b) *If I don't sleep well at night, then I had coffee in the evening.*
 - c) *If I sleep well at night, then I didn't have coffee in the evening.*
 - d) *If I don't sleep well at night, then I didn't have coffee in the evening.*
 - e) *None of the answers is correct*
- 7) What is the inverse of the conditional statement " *I don't sleep well at night whenever I had coffee in the evening*"
- a) *If I didn't had coffee in the evening, then I sleep well at night.*
 - b) *If I had coffee in the evening, then I sleep well at night.*
 - c) *If I sleep well at night, then I didn't have coffee in the evening.*
 - d) *If I don't sleep well at night, then I didn't have coffee in the evening.*
 - e) *None of the answers is correct*
- 8) Determine whether this *biconditional* is true or false.
 $5^2 > 0$ if and only if $0^2 > 0$. [T / F]

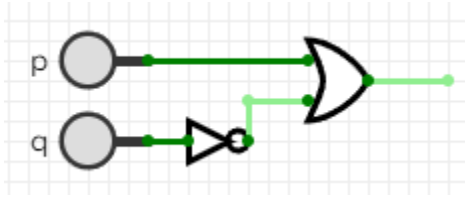
9) Find the output of the following combinatorial circuits.



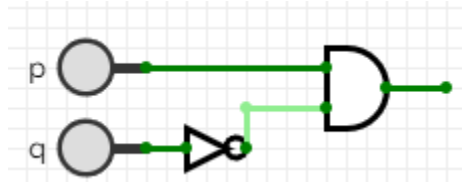
- a) $[(\neg p \wedge r) \vee \neg(p \vee \neg q)] \wedge [\neg(p \vee \neg q) \wedge (\neg r \vee \neg q)]$
- b) $[(\neg p \wedge r) \vee \neg(p \vee \neg q)] \wedge [(p \vee \neg q) \wedge (\neg r \vee \neg q)]$
- c) $[(\neg p \wedge r) \vee (p \vee \neg q)] \wedge \neg[(p \vee \neg q) \wedge (\neg r \vee \neg q)]$
- d) $[(\neg p \wedge r) \vee \neg(p \vee \neg q)] \wedge \neg[\neg(p \vee \neg q) \wedge (\neg r \vee \neg q)]$
- e) None of them

10) Construct a combinatorial circuit using *inverters*, *OR gates*, and *AND gates* that produce the output $((p \vee q) \leftrightarrow q)$ from input bits p and q (You can use logical equivalences to get a simplified result).

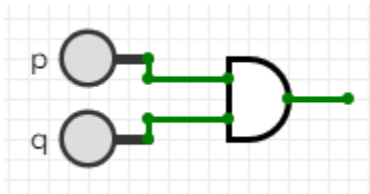
a)



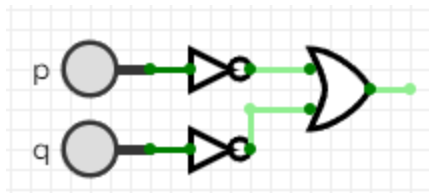
b)



c)



d)



e) None of them

11) Use De Morgan's laws to find the negation of the statement: "*The student will have a class tomorrow or will not have an exam.*"

- a) The student will have a class tomorrow or will have an exam.
- b) The student will not have a class tomorrow and will have an exam.
- c) The student will not have a class tomorrow and will not have an exam.
- d) The student will have a class tomorrow or will not have an exam.
- e) None of the answers is correct

12) Determine whether the following formula is true or false:

$$\neg (p \leftrightarrow q) \equiv \neg p \leftrightarrow \neg q \quad [T / F]$$

13) Determine whether the following formula is true or false:

$$p \rightarrow q \vee r \equiv (p \rightarrow q) \vee (p \rightarrow r) \quad [T / F]$$

14) Determine whether the following formula is true or false:

$$(p \rightarrow q) \wedge (p \rightarrow \neg q) \equiv \neg q \quad [T / F]$$

15) Which one of the following Boolean expressions is NOT a tautology?

- a) $((s \vee q) \wedge (\neg s \vee t)) \rightarrow (q \vee t)$
- b) $((t \rightarrow q) \wedge (q \rightarrow s)) \rightarrow (t \rightarrow s)$
- c) $p \rightarrow (p \vee t)$
- d) $((t \vee q) \wedge \neg t) \rightarrow q$
- e) All of the answers are tautologies.

16) Determine the truth value of the following statement if the domain of each variable consists of all integers and $P(x)$ denotes: " $x^3 = -3$ " ?

$$\exists x P(x) \quad [T / F]$$

17) Determine the truth value of the following statement if the domain of each variable consists of all negative integers $P(x)$ denotes: $x^3 - 2 < -1$?

$$\forall x P(x) \quad [T / F]$$

18) Which one of the following is the most appropriate logical formula to represent the statement? "For any natural number n , n is even iff n^2 is even". The domain is all integer numbers.

The following notations are used:

Even(n): n is even

N(n): n is a natural number

- a) $\forall n (\text{Even}(n^2) \leftrightarrow \text{Even}(n))$
- b) $\forall n (\text{Even}(n) \rightarrow \text{Even}(n^2))$
- c) $\forall n (N(n) \rightarrow (\text{Even}(n) \leftrightarrow \text{Even}(n^2)))$
- d) $\forall n (\text{Even}(n^2) \leftrightarrow \text{Even}(n) \rightarrow N(n))$
- e) All of the answers are incorrect

19) Translate the following statement "All of my friends are not perfect." into logical expressions using predicates, quantifiers, and logical connectives. Consider the predicates: $F(x)$: " x is my friend", and $P(x)$: " x is perfect". Let the domain to be all people.

- a) $\exists x \neg (F(x) \wedge P(x))$
- b) $\neg \exists x (F(x) \wedge P(x))$
- c) $\forall x (\neg P(x) \wedge F(x))$
- d) $\forall x (F(x) \rightarrow \neg P(x))$
- e) All of the answers are incorrect.

20) $P(x)$ is the statement " $x + 1 < 5$ " and the domain consists of the positive integers less than 4. Write out the proposition $\exists x \neg P(x)$ using *disjunctions*, *conjunctions*, and *negations* (i.e. rewrite the propositions without using quantifiers).

- a) $\neg P(1) \wedge \neg P(2) \wedge \neg P(3)$
- b) $\neg P(1) \vee \neg P(2) \vee \neg P(3)$
- c) $\neg P(1) \wedge \neg P(2) \wedge \neg P(3) \wedge \neg P(4)$
- d) $\neg P(1) \vee \neg P(2) \vee \neg P(3) \vee \neg P(4)$

21) $P(x)$ is the statement " $x + 1 < 5$ " and the domain consists of the positive integers less than 4. Write out the proposition $\neg \forall x \neg P(x)$ using *disjunctions*, *conjunctions*, and *negations* (i.e. rewrite the propositions without using quantifiers).

- a) $\neg(\neg P(0) \wedge \neg P(1) \wedge \neg P(2) \wedge \neg P(3))$
- b) $\neg(\neg P(0) \wedge \neg P(1) \wedge \neg P(2) \wedge \neg P(3) \wedge \neg P(4))$
- c) $\neg(\neg P(1) \wedge \neg P(2) \wedge \neg P(3) \wedge \neg P(4))$
- d) $\neg(\neg P(1) \wedge \neg P(2) \wedge \neg P(3))$

22) Select the answers that represent the negation of the following statement: "Some teachers are painters"

[Hint: one or any answers can be selected]

- a) "Everyone are not painters or not teachers"
- b) "Everyone are not painters and not teachers"
- c) "No one is teacher",
- d) "No painters are teachers",